

WEEK 1

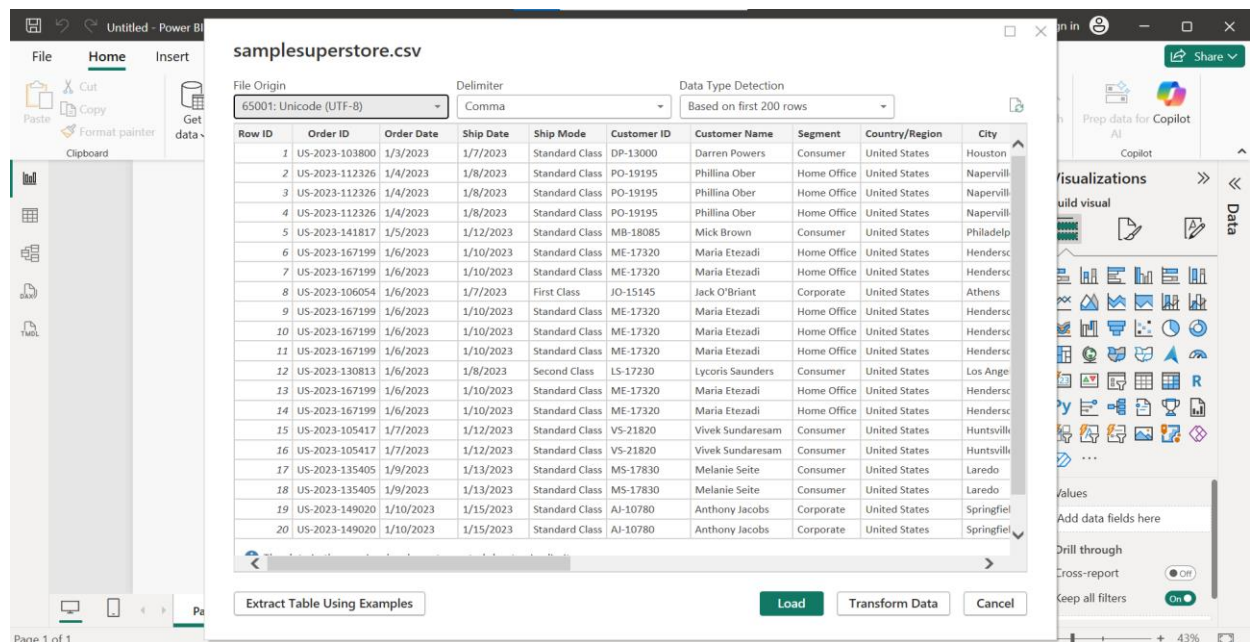
SUPERSTORE SALES DATA IMPORT, POWER QUERY, DATA CLEANING

NAME : BARAKATH SABANA.M

ROLL.NO : E25DA006

1.Data Import

The Superstore Sales dataset was imported into Power BI to begin the data preparation and analysis process. The dataset includes order, customer, shipping, product, and geographical information.



Importing the Superstore Sales dataset into Power BI for further data preparation.

2. Data Type Verification

The imported dataset was opened in Power Query Editor to review the columns and verify their data types. The applied steps show that the headers were promoted and appropriate data types were assigned to the fields.

Untitled - Power Query Editor

File Home Transform Add Column View Tools Help

Close & Apply New Source Recent Sources Enter Data Data source settings Manage Parameters Export query results Refresh Preview Properties Advanced Editor Choose Columns Remove Columns Keep Rows Remove Rows Sort Split Column Group By Data Type: Whole Number Use First Row as Headers Replace Values Merge Queries Append Queries Combine Files Combine

Queries [1]

samplesuperstore

Table.TransformColumnTypes(#"Promoted Headers",{{"Row ID", Int64.Type}, {"Order ID", type text}, {"Order Date", type date}, {"Ship Date", type date}, {"Ship Mode", type text}})

Row ID	Order ID	Order Date	Ship Date	Ship Mode
1	US-2023-103800	1/3/2023	1/7/2023	Standard Class
2	US-2023-112326	1/4/2023	1/8/2023	Standard Class
3	US-2023-112326	1/4/2023	1/8/2023	Standard Class
4	US-2023-112326	1/4/2023	1/8/2023	Standard Class
5	US-2023-141817	1/5/2023	1/12/2023	Standard Class
6	US-2023-167199	1/6/2023	1/10/2023	Standard Class
7	US-2023-167199	1/6/2023	1/10/2023	Standard Class
8	US-2023-106054	1/6/2023	1/7/2023	First Class
9	US-2023-167199	1/6/2023	1/10/2023	Standard Class
10	US-2023-167199	1/6/2023	1/10/2023	Standard Class
11	US-2023-167199	1/6/2023	1/10/2023	Standard Class
12	US-2023-130813	1/6/2023	1/8/2023	Second Class
13	US-2023-167199	1/6/2023	1/10/2023	Standard Class
14	US-2023-167199	1/6/2023	1/10/2023	Standard Class
15	US-2023-105417	1/7/2023	1/12/2023	Standard Class
16	US-2023-105417	1/7/2023	1/12/2023	Standard Class
17	US-2023-135405	1/9/2023	1/13/2023	Standard Class
18	US-2023-135405	1/9/2023	1/13/2023	Standard Class

21 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows

PREVIEW DOWNLOADED AT 21:34

Reviewing column data types and applied transformation steps in Power Query Editor.

3. Date Format Standardization

The date columns were transformed into a consistent date format using Power Query. The Changed Type with Locale option was applied to ensure that the date values are correctly recognized and interpreted.

Untitled - Power Query Editor

File Home Transform Add Column View Tools Help

Close & Apply New Source Recent Sources Enter Data Data source settings Manage Parameters Export query results Refresh Preview Properties Advanced Editor Choose Columns Remove Columns Keep Rows Remove Rows Sort Split Column Group By Data Type: Date Use First Row as Headers Replace Values Merge Queries Append Queries Combine Files Combine

Queries [1]

samplesuperstore

Table.TransformColumnTypes(#"Changed Type with Locale",{{"Ship Date", type date}},{"en-US"})

ID	Order ID	Order Date	Ship Date	Ship Mode	Cus
1	US-2023-103800	03-01-2023	07-01-2023	Standard Class	DP-1
2	US-2023-112326	04-01-2023	08-01-2023	Standard Class	PO-1
3	US-2023-112326	04-01-2023	08-01-2023	Standard Class	PO-1
4	US-2023-112326	04-01-2023	08-01-2023	Standard Class	PO-1
5	US-2023-141817	05-01-2023	12-01-2023	Standard Class	MB-1
6	US-2023-167199	06-01-2023	10-01-2023	Standard Class	ME-1
7	US-2023-167199	06-01-2023	10-01-2023	Standard Class	ME-1
8	US-2023-106054	06-01-2023	07-01-2023	First Class	JO-1
9	US-2023-167199	06-01-2023	10-01-2023	Standard Class	ME-1
10	US-2023-167199	06-01-2023	10-01-2023	Standard Class	ME-1
11	US-2023-167199	06-01-2023	10-01-2023	Standard Class	ME-1
12	US-2023-130813	06-01-2023	08-01-2023	Second Class	LS-1
13	US-2023-167199	06-01-2023	10-01-2023	Standard Class	ME-1
14	US-2023-167199	06-01-2023	10-01-2023	Standard Class	ME-1
15	US-2023-105417	07-01-2023	12-01-2023	Standard Class	VS-2
16	US-2023-105417	07-01-2023	12-01-2023	Standard Class	VS-2
17	US-2023-135405	09-01-2023	13-01-2023	Standard Class	MS-1
18	US-2023-135405	09-01-2023	13-01-2023	Standard Class	MS-1

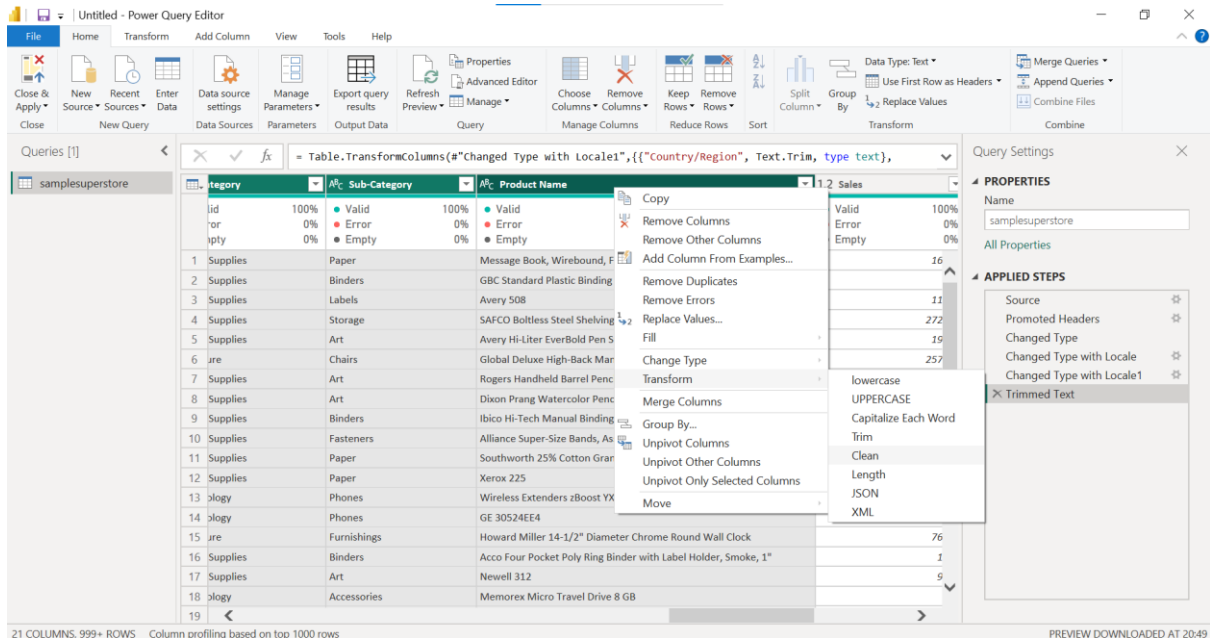
21 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows

PREVIEW DOWNLOADED AT 20:29

Standardizing date columns using locale-based date formatting in Power Query.

4. Text Transformation

The Country/Region column was selected in Power Query, and text transformation options were explored to clean and standardize the text values. These transformations help maintain consistent formatting across the dataset.



Standardizing categorical text fields using Trim and Clean transformations.

5. Year Column Creation

A new Year column was created from the Order Date field using the Date transformation options in Power Query. This allows the sales data to be grouped and analyzed based on different years.

21 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows

PREVIEW DOWNLOADED AT 20:49

Creating a Year column from Order Date for yearly sales analysis.

6. Quarter Column Creation

A new Quarter column was created from the Order Date field using the Date transformation options in Power Query. This helps organize the sales data into quarterly periods for analysis and reporting.

22 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows

PREVIEW DOWNLOADED AT 20:50

Creating a Quarter column from Order Date for quarterly sales analysis.

7. Month Column Creation

A new Month column was created from the Order Date field using the Date transformation options in Power Query. This helps organize and analyze sales data based on monthly periods.

The screenshot shows the Power Query Editor interface. The main area displays a table with columns: Row ID, Order ID, and Order Date. The Order Date column is selected, and a context menu is open, showing options like Year, Month, Quarter, Week, Day, etc. The 'Month' option is highlighted. The right sidebar shows the 'Query Settings' pane with the 'APPLIED STEPS' list, which includes 'Inserted Year' and 'Inserted Quarter'.

Row ID	Order ID	Order Date
1	US-2023-103800	03-01-2023
2	US-2023-112326	04-01-2023
3	US-2023-112326	04-01-2023
4	US-2023-112326	04-01-2023
5	US-2023-141817	05-01-2023
6	US-2023-167199	06-01-2023
7	US-2023-167199	06-01-2023
8	US-2023-106054	06-01-2023
9	US-2023-167199	06-01-2023
10	US-2023-167199	06-01-2023
11	US-2023-167199	06-01-2023
12	US-2023-130813	06-01-2023
13	US-2023-167199	06-01-2023
14	US-2023-167199	06-01-2023
15	US-2023-105417	07-01-2023
16	US-2023-105417	07-01-2023
17	US-2023-135405	09-01-2023
18	US-2023-135405	09-01-2023

Creating a Month column from Order Date for monthly sales analysis.

8. Day Column Creation

A new Day column was created from the Order Date field using the Date transformation options in Power Query. This helps analyze sales data at the daily level and supports detailed time-based analysis.

24 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows

PREVIEW DOWNLOADED AT 20:53

Creating a Day column from Order Date for daily sales analysis.

9. Day Name Column Creation

A new Day Name column was created from the Order Date field using the Date transformation options in Power Query. This converts the date information into weekday names such as Monday, Tuesday, and Wednesday, making it easier to analyze sales by day of the week.

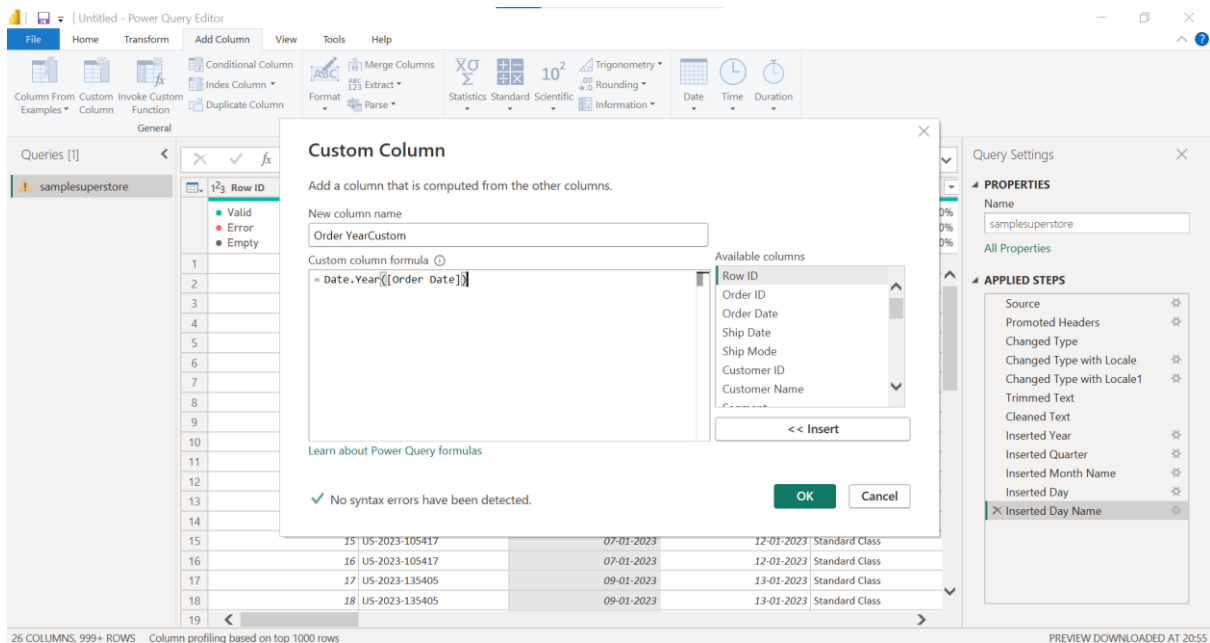
25 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows

PREVIEW DOWNLOADED AT 20:54

Creating a Day Name column from Order Date for weekday-based sales analysis.

10. Custom Column Creation

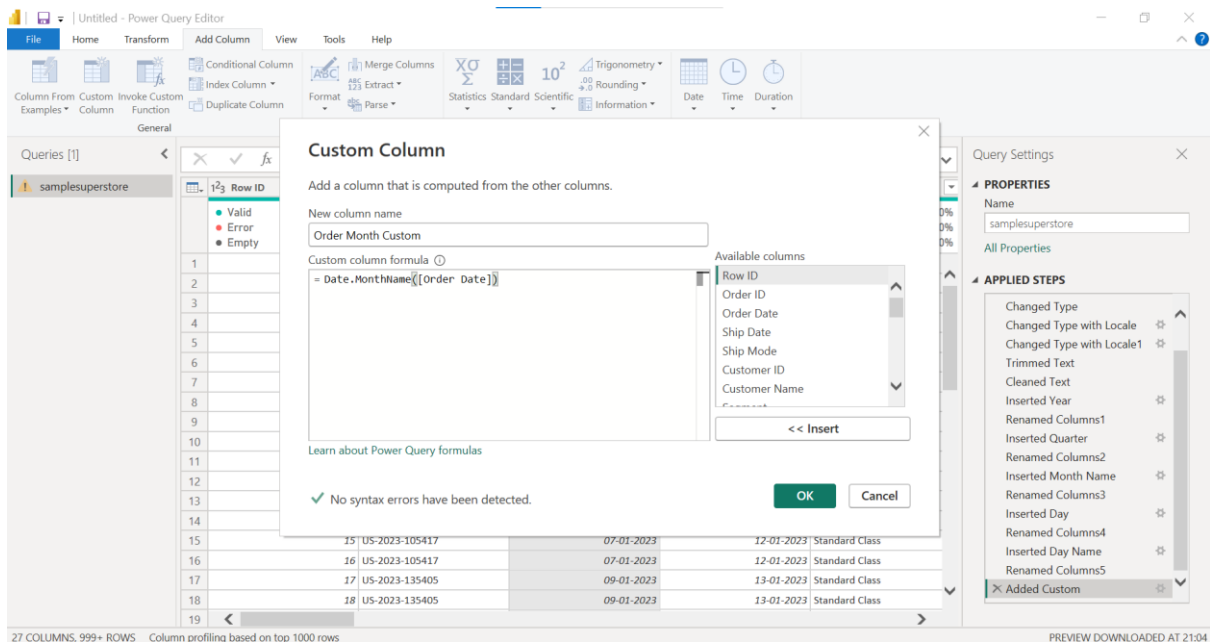
A Custom Column was created in Power Query using a formula based on the Order Date field. The custom column feature allows calculated values to be generated from existing columns and supports further data transformation.



Creating a custom column using the Order Date field in Power Query.

11. Custom Month Column Creation

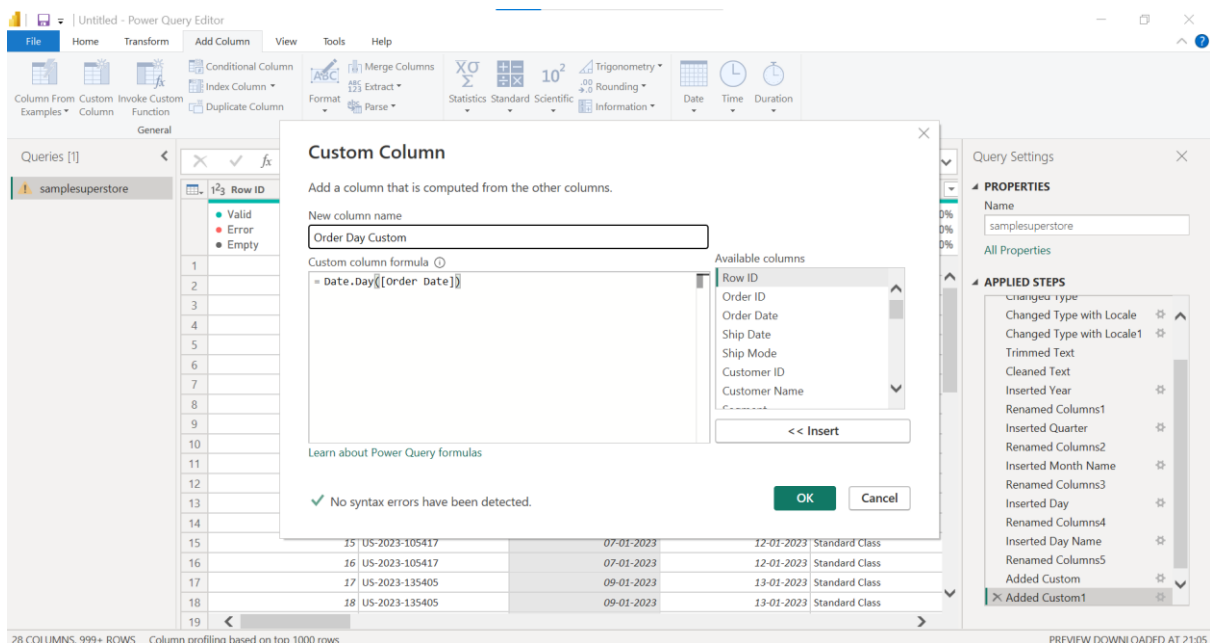
A custom column named Order Month Custom was created using the Date.MonthName([Order Date]) formula. This extracts the month name from the Order Date and makes the data easier to analyze by month.



Creating a custom month-name column from Order Date using Power Query.

12. Custom Day Column Creation

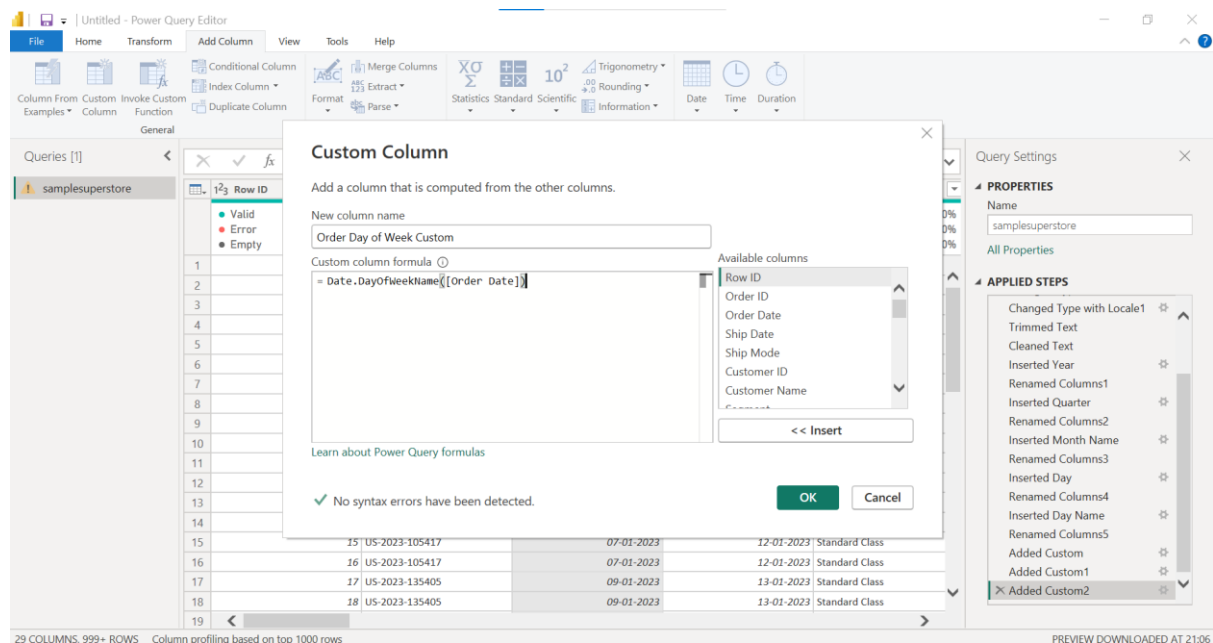
A custom column named Order Day Custom was created using the `Date.Day([Order Date])` formula. This extracts the day value from the Order Date field and supports detailed daily-level analysis.



Creating a custom day column from Order Date using Power Query.

13. Custom Day of Week Column Creation

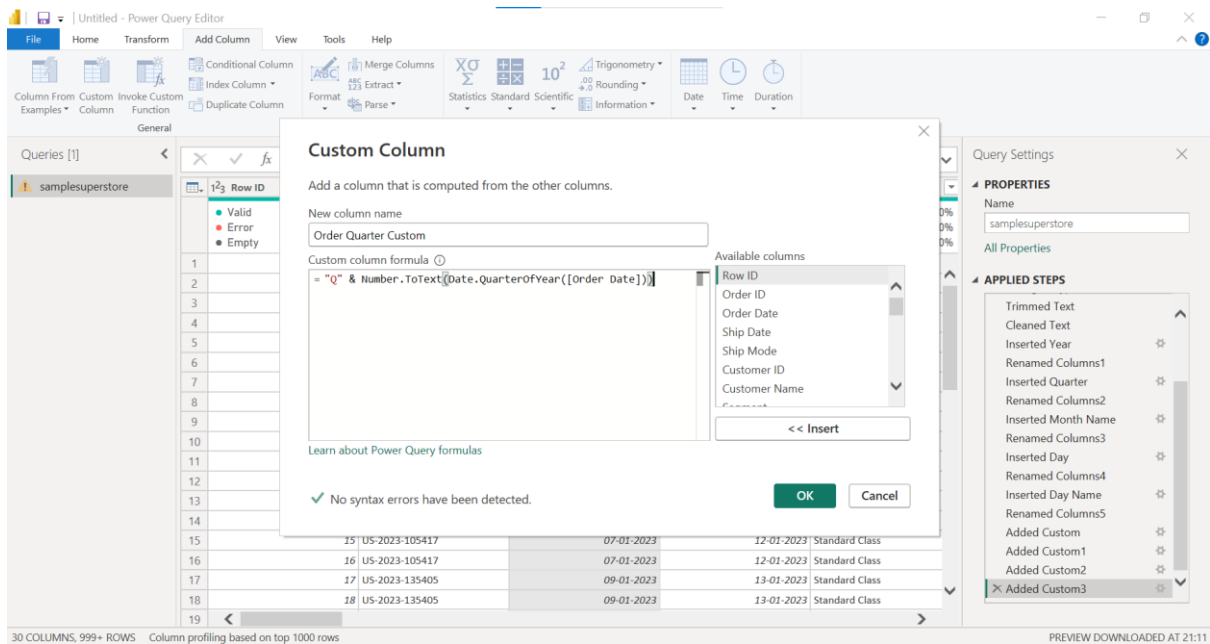
A custom column named Order Day of Week Custom was created using the `Date.DayOfWeek([Order Date])` formula. This extracts the day-of-week value from the Order Date field and supports weekday-based analysis.



Creating a custom day-of-week column from Order Date using Power Query.

14. Custom Quarter Column Creation

A custom column named Order Quarter Custom was created using a Power Query formula based on the Order Date field. The formula calculates the quarter of the year, allowing the sales data to be organized and analyzed by quarterly periods.



Creating a custom quarter column from Order Date using Power Query.

15. Applied Steps Review

The Applied Steps section in Power Query was reviewed to verify the sequence of data transformations. It shows steps such as inserting Year, Quarter, Month Name, Day, and Day Name columns, along with custom columns added during the data-cleaning process.

▲ APPLIED STEPS

Source	⚙
Promoted Headers	⚙
Changed Type	
Changed Type with Locale	⚙
Changed Type with Locale1	⚙
Trimmed Text	
Cleaned Text	
Inserted Year	⚙
Renamed Columns1	
✕ Inserted Quarter	⚙
Renamed Columns2	
Inserted Month Name	⚙
Renamed Columns3	
Inserted Day	⚙
Renamed Columns4	
Inserted Day Name	⚙

Renamed Columns5	
Added Custom	⚙
Added Custom1	⚙
Added Custom2	⚙
Added Custom3	⚙
✕ Added Custom4	⚙

Reviewing the applied transformation steps used to prepare the Superstore Sales dataset.